Mini SIEM Capstone Project – Week 8 Progress Report

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Project Overview

The goal of this project is to design and implement a simplified Security Information and Event Management (SIEM) system capable of ingesting log data, normalizing events into a unified schema, storing them in a database, and performing detection and correlation across multiple data sources. The long-term objective is to simulate real-world SIEM functionality, including identifying and validating malicious activity using labeled datasets.

Summary of Week 8 Work

During Week 8, the project expanded from basic two-source correlation to full multi-source correlation using the LANL cybersecurity dataset. In addition to the previously integrated authentication and red team data, three additional log sources were incorporated:

- proc.txt (process execution events)
- flows.txt (network flow data)
- dns.txt (DNS lookup activity)

This expansion enabled the system to correlate behavior across five different data sources, significantly increasing the realism and analytical capability of the SIEM.

Implementation Details

1. Additional LANL Data Source Integration

Three new parsers were developed to support additional LANL log formats:

- lanl_proc.py for process execution events
- lanl_flows.py for network traffic data
- lanl_dns.py for DNS lookup activity

Each parser extracted relevant fields and mapped them into the existing NormalizedEvent schema. This ensured consistency across all data sources while preserving important attributes such as timestamps, users, computers, and activity types.

Each dataset was sampled to 20,000 records to maintain performance while still providing meaningful data for analysis.

2. Multi-Source Database Integration
All five LANL data sources were successfully ingested and stored within the SQLite database:

- 20,000 authentication events
- 20,000 process events
- 20,000 network flow events
- 20,000 DNS events
- 749 red team attack events

This resulted in a total of over 80,000 normalized events available for correlation and analysis.

3. Multi-Source Correlation Engine
A new correlation script (lanl_multi_source_correlation.py) was developed to analyze relationships across all five datasets. This script performs joins across shared attributes such as destination computers, enabling the system to identify connections between normal activity and known attacker behavior.

The system now supports correlation across:

- Authentication events (who logged in)
- Process execution (what ran)
- Network flows (where traffic was sent)
- DNS activity (what systems were resolved)
- Red team activity (confirmed attacker behavior)

Multi-Source Correlation Results

Multi-Source Correlation Output:

```
Redteam destination overlap:
Auth matches: 49818
Process matches: 6668
Flow matches: 29007
DNS matches: 42947
```

The correlation results demonstrate significant overlap between red team activity and other system behaviors:

- Authentication matches: 49,818
- Process matches: 6,668
- Flow matches: 29,007
- DNS matches: 42,947

Sample Multi-Source Correlated Events:

```
Sample multi-source overlaps:
('150885', 'U620@DOM1', 'C17693', 'C1003', 'lanl_logoff', 'U320', '15052', 'lanl_process_start', 'P8', '266')
('150885', 'U620@DOM1', 'C17693', 'C1003', 'lanl_logoff', 'U320', '15052', 'lanl_process_start', 'P184', '267')
('150885', 'U620@DOM1', 'C17693', 'C1003', 'lanl_logoff', 'U320', '15052', 'lanl_process_start', 'P47', '367')
('150885', 'U620@DOM1', 'C17693', 'C1003', 'lanl_logoff', 'U320', '15052', 'lanl_process_stop', 'P184', '268')
('150885', 'U620@DOM1', 'C17693', 'C1003', 'lanl_logoff', 'U320', '15052', 'lanl_process_stop', 'P47', '373')
('150885', 'U620@DOM1', 'C17693', 'C1003', 'lanl_logoff', 'U629', '15091', 'lanl_process_start', 'P8', '266')
('150885', 'U620@DOM1', 'C17693', 'C1003', 'lanl_logoff', 'U629', '15091', 'lanl_process_start', 'P184', '267')
('150885', 'U620@DOM1', 'C17693', 'C1003', 'lanl_logoff', 'U629', '15091', 'lanl_process_start', 'P47', '367')
('150885', 'U620@DOM1', 'C17693', 'C1003', 'lanl_logoff', 'U629', '15091', 'lanl_process_stop', 'P184', '268')
('150885', 'U620@DOM1', 'C17693', 'C1003', 'lanl_logoff', 'U629', '15091', 'lanl_process_stop', 'P47', '373')
```

Example correlated events demonstrating relationships between red team activity, authentication events, and process execution on shared systems.

Key Takeaways
The most significant achievement of Week 8 is the successful implementation of full multi-source correlation. The system now moves beyond simple event matching and begins to reconstruct behavioral activity across systems.

This enables the SIEM to answer more advanced questions such as:

- What activity occurred on systems targeted by attackers?
- What processes were executed following authentication events?
- What network behavior followed potential compromise?

These capabilities closely mirror real-world SIEM functionality and represent a major step forward in the project.

Challenges Encountered
Several challenges were encountered during this phase:

- Handling multiple large datasets simultaneously
- Ensuring consistent normalization across different log formats
- Managing database size and preventing duplicate data insertion
- Maintaining performance while processing multi-source correlations

All issues were resolved while preserving the overall system design.

Next Steps (Week 9 Plan)
The next phase of the project will focus on detection and validation:

- Develop advanced detection rules based on correlated behavior
- Use red team data as ground truth to validate detections
- Measure detection effectiveness (coverage and accuracy)
- Begin refining alert generation and severity scoring

These steps will transform the system from a correlation engine into a fully functional detection system.